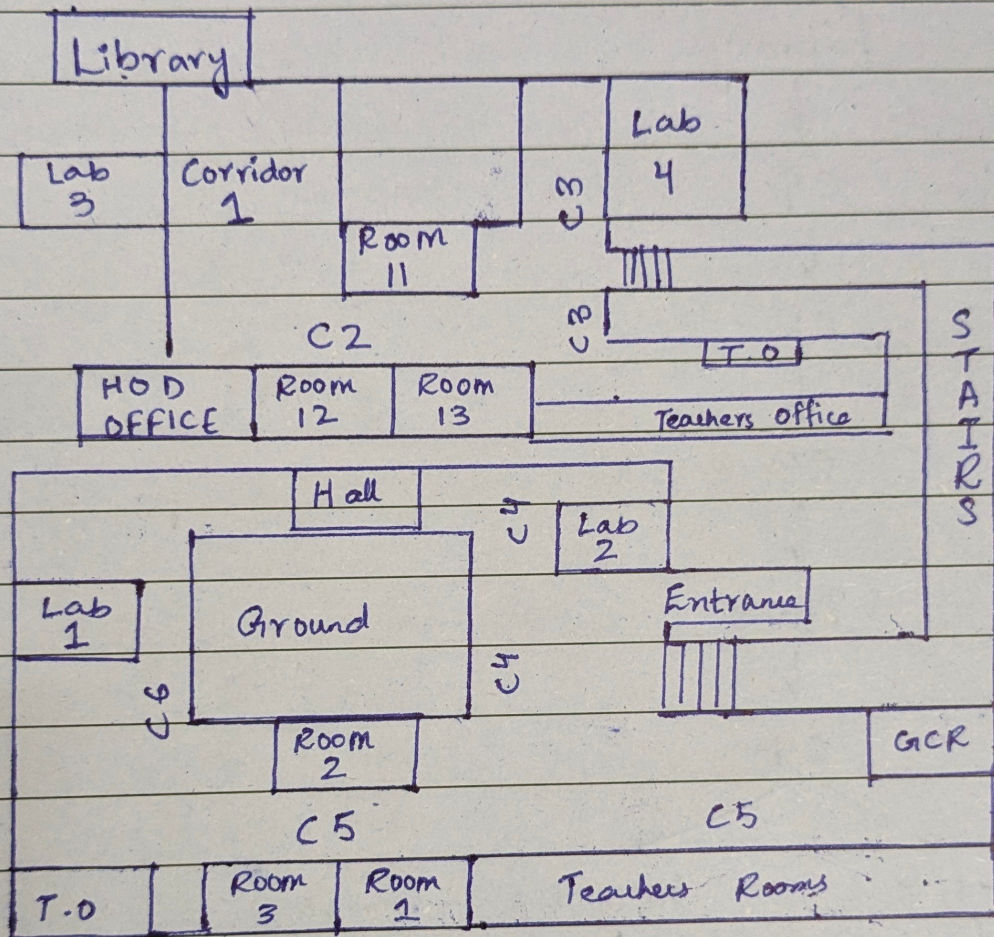


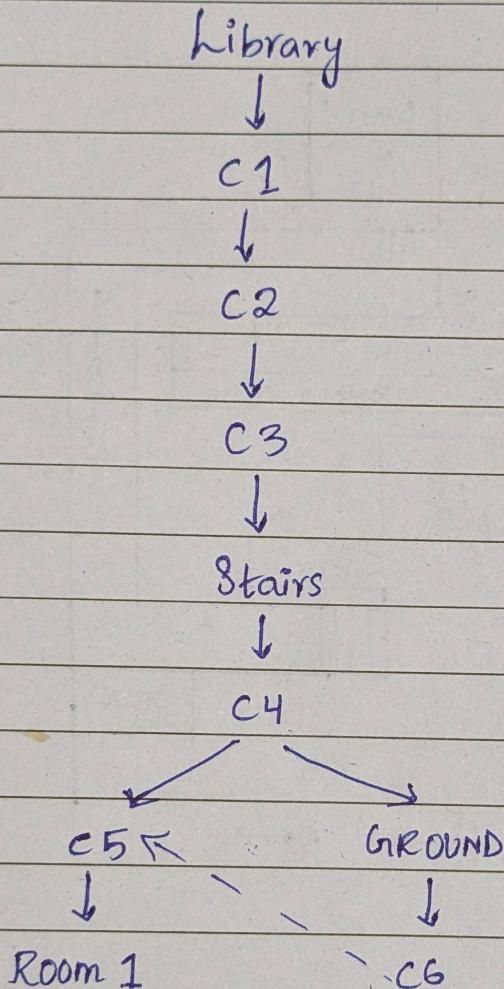
BFS: From Libr. to Room 1

Date _____

• Map :-



. Tree Structure :-



. Graph Structure & Connections :-

Library → ~~Library~~, c2, c1
 c1 → Library, c2
 c2 → c1, c3
 c3 → c2, Stairs

Stairs $\rightarrow$ c3, c4

c4 $\rightarrow$ Stairs, c5, Ground

c5 $\rightarrow$ c4, Room 1, c6

Ground $\rightarrow$ c4, c6

c6 $\rightarrow$ Ground, c5

Room 1 $\rightarrow$ c5

. BFS Trace :-

Library $\rightarrow$ Room 1

Step 0 : Initialization

Q: [Library]

v: [Library]

Step 1: Dequeue Library & Expand

Q: [c1]

v: [Library, c1]

Step 2: Dequeue c1 & Expand

Q: [c2]

v: [Library, c1, c2]

Step 3: Dequeue c2 & Expand

Q: [c3]

v: [Library, c1, c2, c3]

Step 4 : Dequeue C3 & Expand

Q: [Stairs]

V: [Library, c1, c2, c3, Stairs]

Step 5 : Dequeue Stairs & Expand

Q: [C4]

V: [Library, c1, c2, c3, Stairs, C4]

Step 6 : Dequeue C4 & Expand

Q: [C5, Ground]

V: [Library, c1, c2, c3, Stairs, C4, C5, Ground]

Step 7 : Dequeue C5 & Expand

Q: [Ground, Room 1, C6]

V: [Library, c1, c2, c3, Stairs, C4, C5, Ground, Room 1, C6]

Step 8 : Dequeue Ground

Q: [Room 1, C6]

V: [Library, c1, c2, c3, Stairs, C4, C5, Ground, Room 1, C6]

Step 9 : Dequeue Room 1

Goal Found !

Date _____

• BFS ORDER OF EXPANSION:-

Library $\rightarrow$ C1 $\rightarrow$ C2 $\rightarrow$ C3 $\rightarrow$ Stairs $\rightarrow$ C4 $\rightarrow$ C5 $\rightarrow$ Ground $\rightarrow$ Room 1

• SHORTEST PATH FOUND :-

Library $\rightarrow$ C1 $\rightarrow$ C2 $\rightarrow$ C3 $\rightarrow$ Stairs $\rightarrow$ C4 $\rightarrow$ C5 $\rightarrow$ Room 1